DSC550Milestone2Schrotberger-Submitted

August 8, 2026

1 Data Mining Course Project

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2.1 Milestone 1: Data Selection and EDA

Post-secondary education is at a unique point in this economy. With tuition rates rising, universities want to be able to convince students to attend their location. As students search for an institution, things they look at are location, degree programs offered, and cost. With the goal being obtaining a degree at an appropriate cost. Schools become less desirable when there is negative press or they have low or decreasing graduation rates. To ensure that students meet their goals and universities are desirable to the next generation of students, they must maintain their graduation rates.

Performance of students in education can be impacted by many outside factors. Knowing a student's background, initial performance, and economy they come from can help predict if they are likely to graduate. In the secondary education environment, knowing what kinds of variables impact if someone will graduate can really help students and universities alike. Being able to predict who might need assistance to graduate will increase graduation rates and provide students with backing and support. The goal is to maintain or increase graduation rates.

I found a data set on Kaggle that has been published as well. It includes data from 2008-2019 from Europe in 17 undergraduate degree programs. There is demographic, socioeconomic, and economic metrics available. The target variable has the categorical options: Dropout, Enrolled, and Graduate. As the goal of this project is to determine graduation success, those with an Enrolled status will be dropped from the analysis as their true outcome has not been determined yet. There are 4,424 records in the data set.

https://www.kaggle.com/datasets/thedevastator/higher-education-predictors-of-student-retention

[1]:
```python
# Import packages needed for analysis
import numpy as np
import pandas as pd
import warnings
warnings.simplefilter(action='ignore', category=FutureWarning)
import matplotlib.pyplot as plt
import seaborn as sns
```

2.1.1 Import Data

[3]:
```
# Import CSV to dataframe
students=pd.read_csv(r'G:/My Drive/College/Caitie/DSC550-T301/student_dataset.
 ↪csv')
# Display first 5 rows of data
students.head()
```

[3]:
```
   Marital status  Application mode  Application order  Course  \
0               1                 8                  5       2
1               1                 6                  1      11
2               1                 1                  5       5
3               1                 8                  2      15
4               2                12                  1       3

   Daytime/evening attendance  Previous qualification  Nacionality  \
0                           1                       1            1
1                           1                       1            1
2                           1                       1            1
3                           1                       1            1
4                           0                       1            1

   Mother's qualification  Father's qualification  Mother's occupation  …  \
0                      13                      10                    6  …
1                       1                       3                    4  …
2                      22                      27                   10  …
3                      23                      27                    6  …
4                      22                      28                   10  …

   Curricular units 2nd sem (credited)  Curricular units 2nd sem (enrolled)  \
0                                    0                                    0
1                                    0                                    6
2                                    0                                    6
3                                    0                                    6
4                                    0                                    6

   Curricular units 2nd sem (evaluations)  \
0                                       0
1                                       6
2                                       0
3                                      10
4                                       6

   Curricular units 2nd sem (approved)  Curricular units 2nd sem (grade)  \
0                                    0                          0.000000
1                                    6                         13.666667
2                                    0                          0.000000
```

```
3                                    5                        12.400000
4                                    6                        13.000000

   Curricular units 2nd sem (without evaluations)  Unemployment rate  \
0                                               0               10.8
1                                               0               13.9
2                                               0               10.8
3                                               0                9.4
4                                               0               13.9

   Inflation rate   GDP    Target
0             1.4  1.74   Dropout
1            -0.3  0.79  Graduate
2             1.4  1.74   Dropout
3            -0.8 -3.12  Graduate
4            -0.3  0.79  Graduate

[5 rows x 35 columns]
```

2.1.2 Exploring the Data

Do a graphical analysis creating a minimum of four graphs. Label your graphs appropriately and explain/analyze the information provided by each graph. Your analysis should begin to answer the question(s) you are addressing. Write a short overview/conclusion of the insights gained from your graphical analysis.

[5]:
```python
students.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4424 entries, 0 to 4423
Data columns (total 35 columns):
 #   Column                                          Non-Null Count  Dtype
---  ------                                          --------------  -----
 0   Marital status                                  4424 non-null   int64
 1   Application mode                                4424 non-null   int64
 2   Application order                               4424 non-null   int64
 3   Course                                          4424 non-null   int64
 4   Daytime/evening attendance                      4424 non-null   int64
 5   Previous qualification                          4424 non-null   int64
 6   Nacionality                                     4424 non-null   int64
 7   Mother's qualification                          4424 non-null   int64
 8   Father's qualification                          4424 non-null   int64
 9   Mother's occupation                             4424 non-null   int64
 10  Father's occupation                             4424 non-null   int64
 11  Displaced                                       4424 non-null   int64
 12  Educational special needs                       4424 non-null   int64
 13  Debtor                                          4424 non-null   int64
 14  Tuition fees up to date                         4424 non-null   int64
```

```
 15  Gender                                      4424 non-null   int64
 16  Scholarship holder                          4424 non-null   int64
 17  Age at enrollment                           4424 non-null   int64
 18  International                               4424 non-null   int64
 19  Curricular units 1st sem (credited)         4424 non-null   int64
 20  Curricular units 1st sem (enrolled)         4424 non-null   int64
 21  Curricular units 1st sem (evaluations)      4424 non-null   int64
 22  Curricular units 1st sem (approved)         4424 non-null   int64
 23  Curricular units 1st sem (grade)            4424 non-null   float64
 24  Curricular units 1st sem (without evaluations) 4424 non-null   int64
 25  Curricular units 2nd sem (credited)         4424 non-null   int64
 26  Curricular units 2nd sem (enrolled)         4424 non-null   int64
 27  Curricular units 2nd sem (evaluations)      4424 non-null   int64
 28  Curricular units 2nd sem (approved)         4424 non-null   int64
 29  Curricular units 2nd sem (grade)            4424 non-null   float64
 30  Curricular units 2nd sem (without evaluations) 4424 non-null   int64
 31  Unemployment rate                           4424 non-null   float64
 32  Inflation rate                              4424 non-null   float64
 33  GDP                                         4424 non-null   float64
 34  Target                                      4424 non-null   object
dtypes: float64(5), int64(29), object(1)
memory usage: 1.2+ MB
```

The data set has 4,424 records and 35 columns. Most of the fields are numeric (int64 or float64) with the exception of the target variable which is currently of object format.

[7]:
```
# Removing "Enrolled" target rows
students_data = students[students['Target'] !='Enrolled']
students_data.info()
# The new dataframe has 3,630 rows: 2,209 Graduate and 1,421 Dropouts
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 3630 entries, 0 to 4423
Data columns (total 35 columns):
 #   Column                                      Non-Null Count  Dtype
---  ------                                      --------------  -----
 0   Marital status                              3630 non-null   int64
 1   Application mode                            3630 non-null   int64
 2   Application order                           3630 non-null   int64
 3   Course                                      3630 non-null   int64
 4   Daytime/evening attendance                  3630 non-null   int64
 5   Previous qualification                      3630 non-null   int64
 6   Nacionality                                 3630 non-null   int64
 7   Mother's qualification                      3630 non-null   int64
 8   Father's qualification                      3630 non-null   int64
 9   Mother's occupation                         3630 non-null   int64
 10  Father's occupation                         3630 non-null   int64
 11  Displaced                                   3630 non-null   int64
 12  Educational special needs                   3630 non-null   int64
```

```
 13  Debtor                                          3630 non-null   int64
 14  Tuition fees up to date                         3630 non-null   int64
 15  Gender                                          3630 non-null   int64
 16  Scholarship holder                              3630 non-null   int64
 17  Age at enrollment                               3630 non-null   int64
 18  International                                   3630 non-null   int64
 19  Curricular units 1st sem (credited)             3630 non-null   int64
 20  Curricular units 1st sem (enrolled)             3630 non-null   int64
 21  Curricular units 1st sem (evaluations)          3630 non-null   int64
 22  Curricular units 1st sem (approved)             3630 non-null   int64
 23  Curricular units 1st sem (grade)                3630 non-null   float64
 24  Curricular units 1st sem (without evaluations)  3630 non-null   int64
 25  Curricular units 2nd sem (credited)             3630 non-null   int64
 26  Curricular units 2nd sem (enrolled)             3630 non-null   int64
 27  Curricular units 2nd sem (evaluations)          3630 non-null   int64
 28  Curricular units 2nd sem (approved)             3630 non-null   int64
 29  Curricular units 2nd sem (grade)                3630 non-null   float64
 30  Curricular units 2nd sem (without evaluations)  3630 non-null   int64
 31  Unemployment rate                               3630 non-null   float64
 32  Inflation rate                                  3630 non-null   float64
 33  GDP                                             3630 non-null   float64
 34  Target                                          3630 non-null   object
dtypes: float64(5), int64(29), object(1)
memory usage: 1020.9+ KB
```

[9]:
```python
# Add column where Graduate = 1 and Dropout = 0
students_data['Target_Int'] = students_data['Target'].apply(lambda x: 1 if x ==
 ↪'Graduate' else 0)
# Update column names to remove '
students_data.rename(columns={"Father's qualification": "Dad qualification"},
 ↪inplace=True)
students_data.rename(columns={"Mother's qualification": "Mom qualification"},
 ↪inplace=True)
students_data.rename(columns={"Father's occupation": "Dad occupation"},
 ↪inplace=True)
students_data.rename(columns={"Mother's occupation": "Mom occupation"},
 ↪inplace=True)
students_data.rename(columns={"Daytime/evening attendance": "Attendance Time"},
 ↪inplace=True)
students_data.head()
```

```
C:\Users\caiti\AppData\Local\Temp\ipykernel_27084\2564270684.py:2:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
```

```
  students_data['Target_Int'] = students_data['Target'].apply(lambda x: 1 if x
== 'Graduate' else 0)
C:\Users\caiti\AppData\Local\Temp\ipykernel_27084\2564270684.py:4:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  students_data.rename(columns={"Father's qualification": "Dad qualification"},
inplace=True)
C:\Users\caiti\AppData\Local\Temp\ipykernel_27084\2564270684.py:5:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  students_data.rename(columns={"Mother's qualification": "Mom qualification"},
inplace=True)
C:\Users\caiti\AppData\Local\Temp\ipykernel_27084\2564270684.py:6:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  students_data.rename(columns={"Father's occupation": "Dad occupation"},
inplace=True)
C:\Users\caiti\AppData\Local\Temp\ipykernel_27084\2564270684.py:7:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  students_data.rename(columns={"Mother's occupation": "Mom occupation"},
inplace=True)
C:\Users\caiti\AppData\Local\Temp\ipykernel_27084\2564270684.py:8:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  students_data.rename(columns={"Daytime/evening attendance": "Attendance
Time"}, inplace=True)
```

[9]:

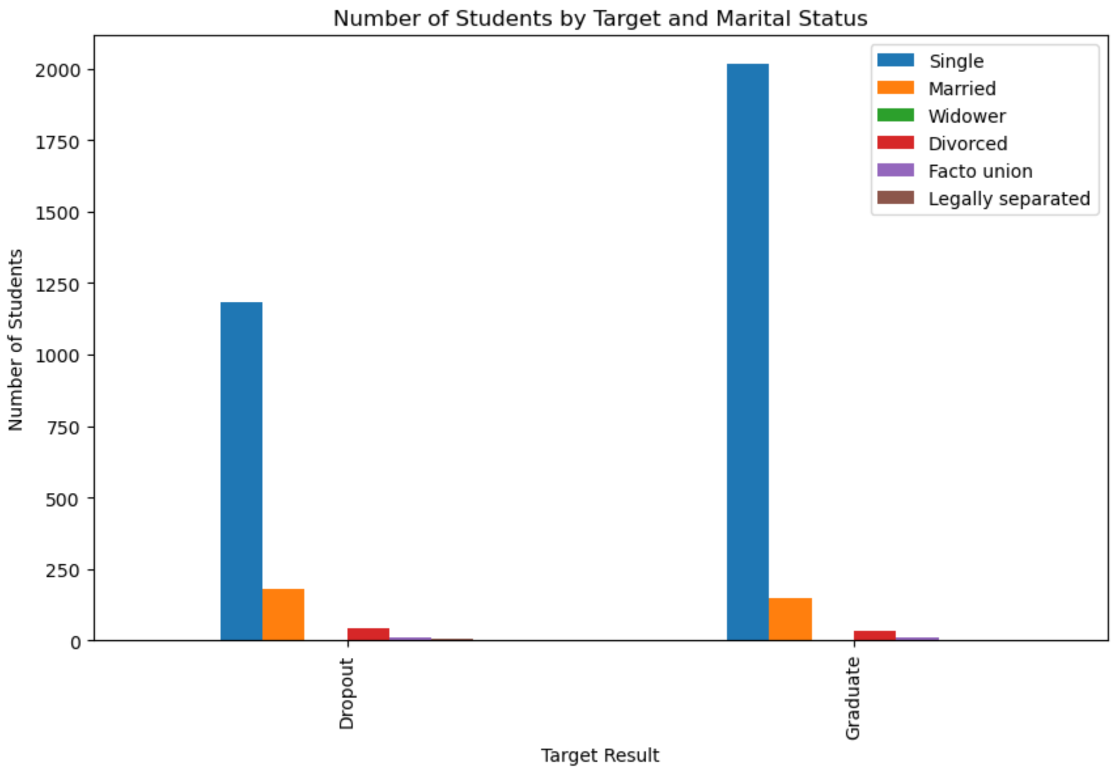
```
   Marital status  Application mode  Application order  Course  \
0               1                 8                  5       2
1               1                 6                  1      11
2               1                 1                  5       5
```

```
3               1                 8                  2      15
4               2                12                  1       3

   Attendance Time  Previous qualification  Nacionality  Mom qualification  \
0                1                       1            1                 13
1                1                       1            1                  1
2                1                       1            1                 22
3                1                       1            1                 23
4                0                       1            1                 22

   Dad qualification  Mom occupation  …  \
0                 10               6  …
1                  3               4  …
2                 27              10  …
3                 27               6  …
4                 28              10  …

   Curricular units 2nd sem (enrolled)  \
0                                    0
1                                    6
2                                    6
3                                    6
4                                    6

   Curricular units 2nd sem (evaluations)  \
0                                       0
1                                       6
2                                       0
3                                      10
4                                       6

   Curricular units 2nd sem (approved)  Curricular units 2nd sem (grade)  \
0                                    0                          0.000000
1                                    6                         13.666667
2                                    0                          0.000000
3                                    5                         12.400000
4                                    6                         13.000000

   Curricular units 2nd sem (without evaluations)  Unemployment rate  \
0                                               0               10.8
1                                               0               13.9
2                                               0               10.8
3                                               0                9.4
4                                               0               13.9

   Inflation rate   GDP    Target  Target_Int
0             1.4  1.74   Dropout           0
```

```
1              -0.3  0.79  Graduate          1
2               1.4  1.74   Dropout          0
3              -0.8 -3.12  Graduate          1
4              -0.3  0.79  Graduate          1

[5 rows x 36 columns]
```

2.1.3 Visualization One

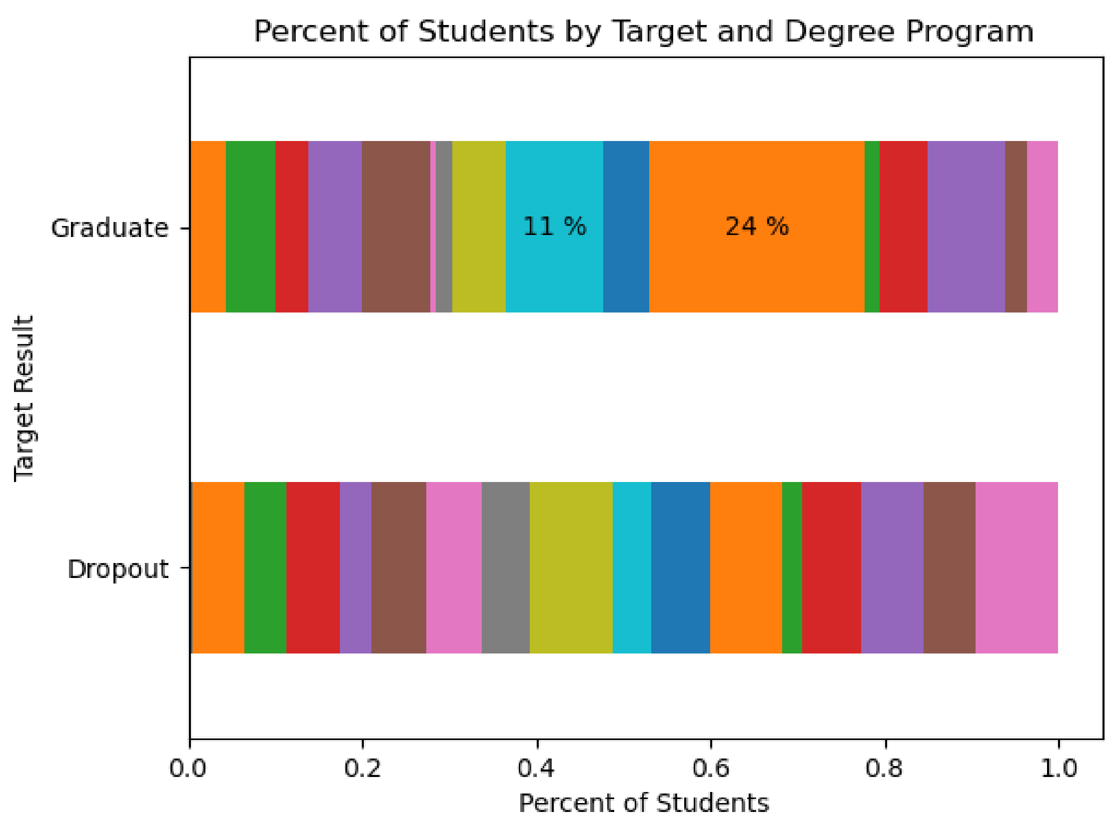
```
[11]: # Graph 1: Student marital status by Target
pivot_table = students_data.pivot_table(index='Target', columns='Marital␣
↪status', values='Target_Int', aggfunc='count')
pivot_table.plot(kind='bar', figsize=(10, 6))
plt.title('Number of Students by Target and Marital Status')
plt.xlabel('Target Result')
plt.ylabel('Number of Students')
new_labels1 = ['Single', 'Married', 'Widower', 'Divorced', 'Facto union',␣
↪'Legally separated']
plt.legend(labels=new_labels1)
plt.show()
```



Based on this graph, the majority of students are single, with a greater proportion who graduate being single as well.

2.1.4 Visualization Two

[13]:
```
# Graph 2:  Percent of students by course and target
x_var, y_var = "Target", "Course"
df_grouped = students_data.groupby(x_var)[y_var].value_counts(normalize=True).
 ↪unstack(y_var)
df_grouped.plot.barh(stacked=True, legend=False)
for ix, row in df_grouped.reset_index(drop=True).iterrows():
#    print(ix, row)
    cumulative = 0
    for element in row:
        if element == element and element > 0.1:
            plt.text(
                cumulative + element / 2,
                ix,
                f"{int(element * 100)} %",
                va="center",
                ha="center",
            )
        cumulative += element
plt.title('Percent of Students by Target and Degree Program')
plt.xlabel('Percent of Students')
plt.ylabel('Target Result')
plt.show()
```

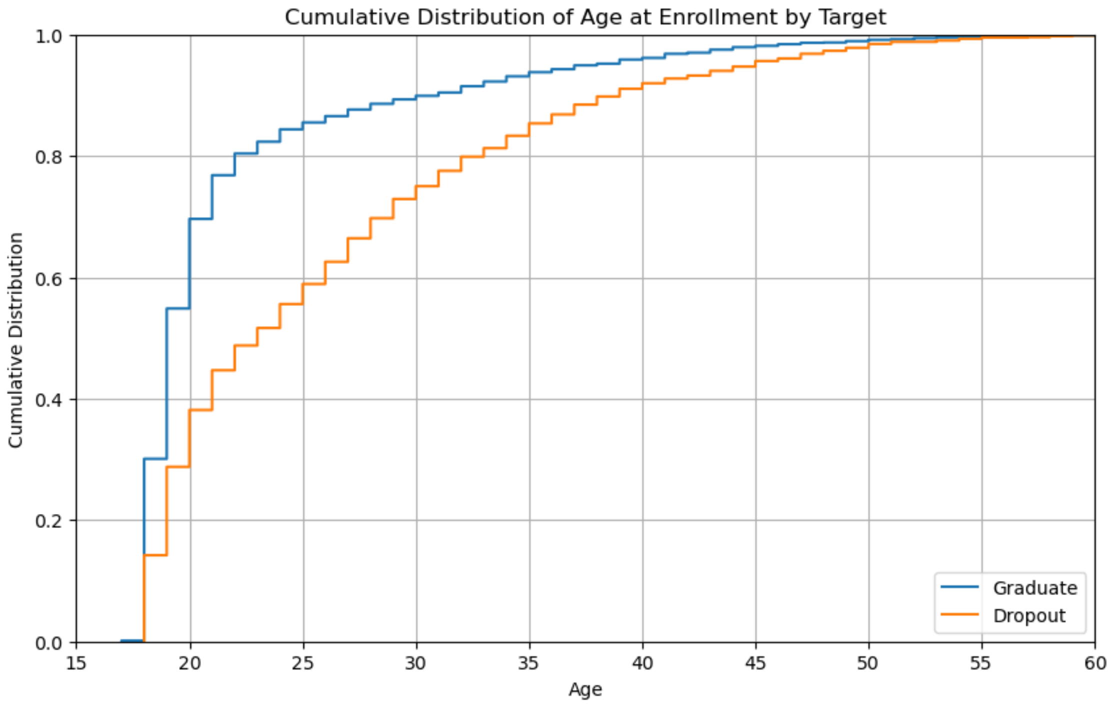


The colors from left to right represent the following course programs, with the total number of students in the programs in (): Biofuel Production Technologies (9), Animation and Multimedia Design (178), Social Service (evening attendance) (194), Agronomy (173), Communication Design(184), Veterinary Nursing (262), Informatics Engineering (106), Equiniculture (120), Management (272), Social Service (313), Tourism (211), Nursing (666), Oral Hygiene (69), Advertising and Marketing Management (220), Journalism and Communication (297), Basic Education (142), Management (evening attendance) (214). Based on the small volumes, Biofuel Production Technologies doesn't visually show on the graph.

Twenty percent of those that graduate are in the Nursing program, followed 11% in Social Services. For those that dropout, the distribution between programs appears more even. This would als0 mean that a greater percentage of students in those programs graduate.

2.1.5 Visualization Three

[15]:
```
#Graph 3:  Cumulative Distribution Functions of Age by Target
student_data_filtered = students_data[['Target','Age at enrollment']]
sorted_multi = student_data_filtered.sort_values(by=['Target', 'Age at␣
↪enrollment'])
sorted_multi = sorted_multi.sort_values(by=['Target', 'Age at enrollment'],␣
↪ascending=[False, True])
# Plot CDF for 'Age at enrollment' split by 'Target'
plt.figure(figsize=(10, 6))
for day in sorted_multi['Target'].unique():
    subset = sorted_multi[sorted_multi['Target'] == day]
    sns.ecdfplot(data=subset, x="Age at enrollment", label=f'{day}')
plt.title('Cumulative Distribution of Age at Enrollment by Target')
plt.xlabel('Age')
plt.ylabel('Cumulative Distribution')
plt.legend(loc='lower right')
plt.xlim(15, 60)
plt.grid(True)
plt.show()
```



This graph shows that the younger a student enrolls in college the more likely they are to graduate. For graduating students, over 50% had enrolled by age 20, while 50% of dropouts had enrolled by age 24. Eighty percent of students who graduated had enrolled by age 23 where the 80% of the dropouts enrolled by age 33. The curve of the dropouts is less sharp initially showing that those that dropout are likely to have started at an older age than those that graduate.

2.1.6 Visualization Four

[17]:
```
# Graph 4:  Target by Mother's Education Level
# The Education levels were very split out so I groupped them togheter for␣
↪analysis purposes

def school_type(x):
    if x in [24, 25, 26, 27]:
        return '1-Elementary'
    elif x in [9, 18, 20, 21,28]:
        return '2-Middle School'
    elif x in [1, 7, 8, 10, 11, 12, 14, 15, 17, 19]:
        return '3-High School'
    elif x in [13, 16, 22, 23, 29, 31, 32]:
        return '4-Associate'
    elif x in [2, 3, 4, 5, 6, 30, 33, 34]:
        return '5-College'
    else:
```

```
        return 'Other'

mom_edu = students_data[['Target','Mom qualification']]
mom_edu['Mom School'] = mom_edu['Mom qualification'].apply(school_type)
mom_edu = mom_edu.sort_values(by=['Target', 'Mom School'], ascending=[False,␣
↪True])
mom_edu.head()
```

```
C:\Users\caiti\AppData\Local\Temp\ipykernel_27084\2359691928.py:19:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  mom_edu['Mom School'] = mom_edu['Mom qualification'].apply(school_type)
```

[17]:

	Target	Mom qualification	Mom School
31	Graduate	27	1-Elementary
225	Graduate	25	1-Elementary
281	Graduate	24	1-Elementary
317	Graduate	26	1-Elementary
336	Graduate	26	1-Elementary

[19]:

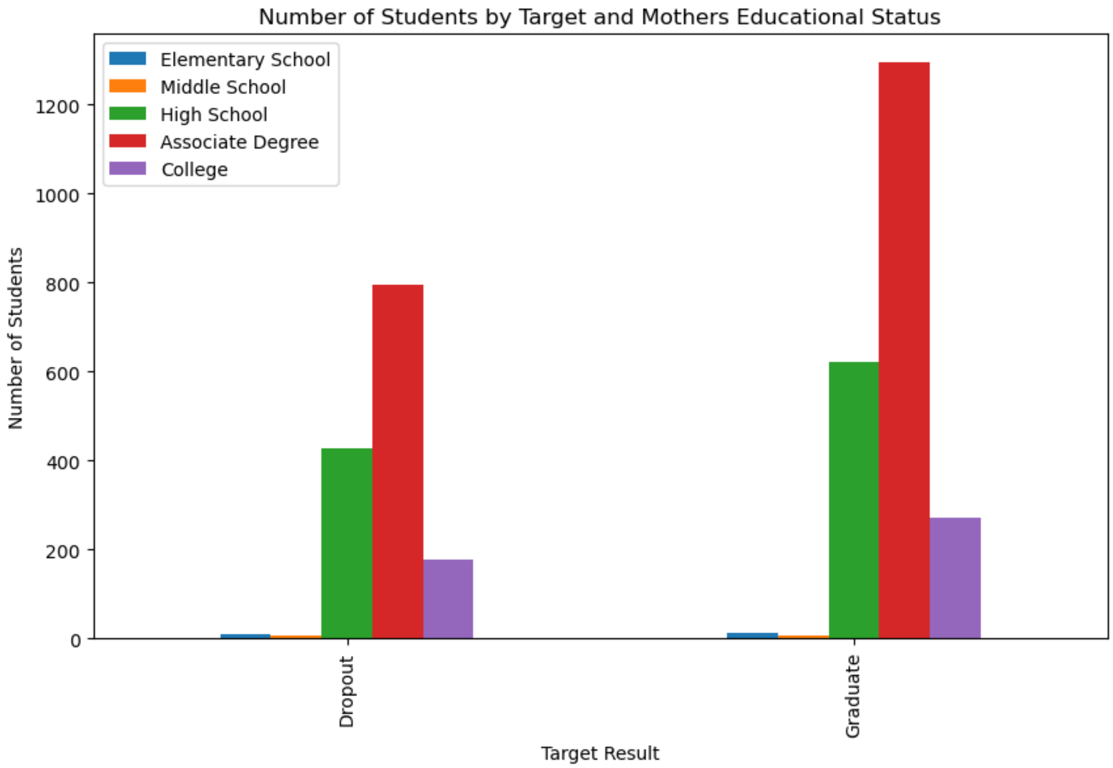
```
# Counts for each target and school level
pivot_table2 = mom_edu.pivot_table(index='Target', columns='Mom School',␣
↪values='Mom qualification', aggfunc='count')
pivot_table2
```

[19]:

Mom School Target	1-Elementary	2-Middle School	3-High School	4-Associate	5-College
Dropout	11	8	428	796	178
Graduate	13	7	622	1294	273

[21]:

```
# Graphing target by mom educational type
pivot_table2.plot(kind='bar', figsize=(10, 6))
plt.title('Number of Students by Target and Mother''s Educational Status')
plt.xlabel('Target Result')
plt.ylabel('Number of Students')
new_labels1 = ['Elementary School', 'Middle School', 'High School', 'Associate␣
↪Degree', 'College', 'Other']
plt.legend(labels=new_labels1)
plt.show()
```



The results between the target and the mother's educational status is very similar. Those that graduate have 70.9% with mothers that had post-secondary education. The dropouts only have 68.5% with mothers who have post-secondary education. This means that the mother's educational background could influence if a student graduates or drops out.

2.1.7 Visualization Five

[23]:
```
# Graph 5:  Target by Father's Education Level
# The Education levels were very split out so I groupped them togheter for␣
↪analysis purposes
dad_edu = students_data[['Target','Dad qualification']]
dad_edu['Dad School'] = dad_edu['Dad qualification'].apply(school_type)
dad_edu = dad_edu.sort_values(by=['Target', 'Dad School'], ascending=[False,␣
↪True])
# Counts for each target and school level
pivot_table3 = dad_edu.pivot_table(index='Target', columns='Dad School',␣
↪values='Dad qualification', aggfunc='count')
pivot_table3
```

```
C:\Users\caiti\AppData\Local\Temp\ipykernel_27084\1142464794.py:4:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead
```

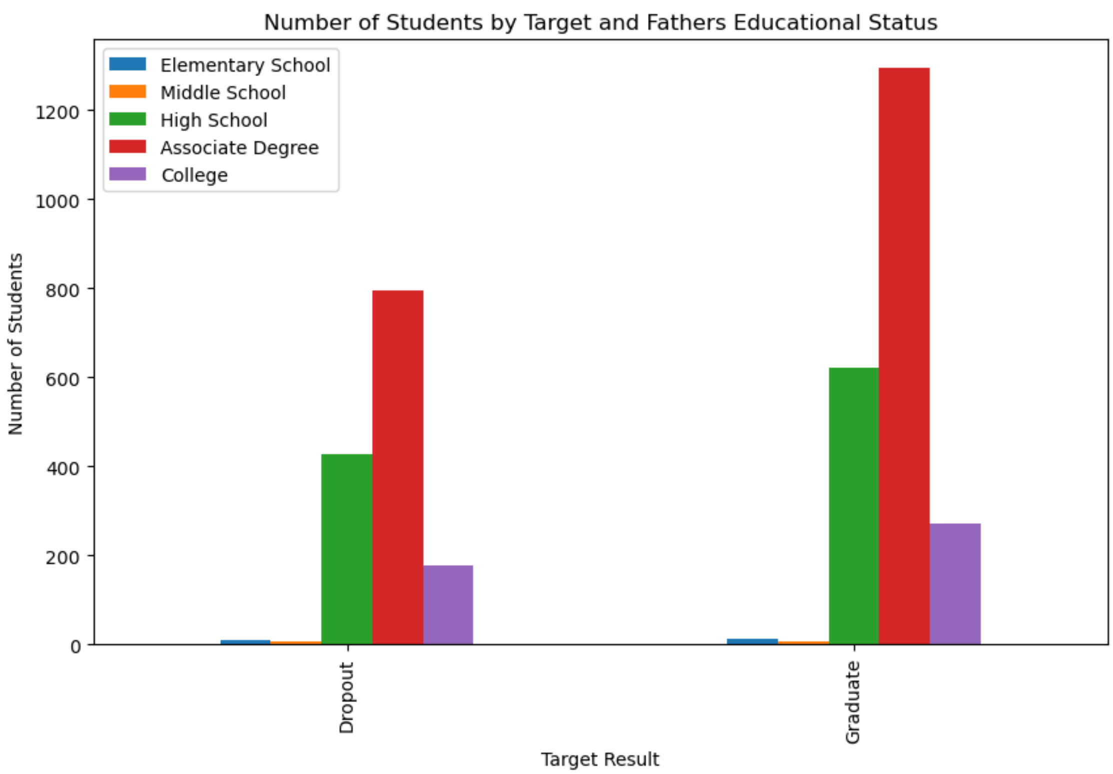
```
See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  dad_edu['Dad School'] = dad_edu['Dad qualification'].apply(school_type)
```

[23]:

```
Dad School  1-Elementary  2-Middle School  3-High School  4-Associate  \
Target
Dropout              520              176            567           16
Graduate             609              417            999            6

Dad School  5-College
Target
Dropout           142
Graduate          178
```

[25]:

```
# Graphing target by mom educational type
pivot_table2.plot(kind='bar', figsize=(10, 6))
plt.title('Number of Students by Target and Father''s Educational Status')
plt.xlabel('Target Result')
plt.ylabel('Number of Students')
new_labels1 = ['Elementary School', 'Middle School', 'High School', 'Associate␣
↪Degree', 'College', 'Other']
plt.legend(labels=new_labels1)
plt.show()
```



The results between the target and the father's educational status is very different - even different from the mother's educational status. Those that graduate have 8.3% with fathers that had post-secondary education. The dropouts only have 2.1% with fathers who have post-secondary education. This means that the father's educational background could significantly influence if a student graduates or drops out. Using both metrics together may work even better together.

2.2 Milestone 2: Data Preparation

2.2.1 Data Preparation done in Milestone 1

Removed Enrolled student data because their data couldn't predict the outcome of Graduate or Dropout

students_data = students[students['Target'] !='Enrolled']

Add column where Graduate = 1 and Dropout = 0 called Target_Int

students_data['Target_Int'] = students_data['Target'].apply(lambda x: 1 if x == 'Graduate' else 0)

Updated column names to remove '

students_data.rename(columns={"Father's qualification": "Dad qualification"}, inplace=True)

students_data.rename(columns={"Mother's qualification": "Mom qualification"}, inplace=True)

students_data.rename(columns={"Father's occupation": "Dad occupation"}, inplace=True)

students_data.rename(columns={"Mother's occupation": "Mom occupation"}, inplace=True)

students_data.rename(columns={"Daytime/evening attendance": "Attendance Time"}, inplace=True)

2.2.2 Review Summary Statistics on Numerical Data

[27]:
```
# Summary statistics of the numerical data
students_data[['Age at enrollment', 'Curricular units 1st sem (credited)',␣
↪'Curricular units 1st sem (credited)',
              'Curricular units 1st sem (enrolled)', 'Curricular units 1st sem␣
↪(evaluations)',
              'Curricular units 1st sem (approved)', 'Curricular units 1st sem␣
↪(grade)',
              'Curricular units 1st sem (without evaluations)', 'Curricular␣
↪units 2nd sem (credited)',
              'Curricular units 2nd sem (enrolled)', 'Curricular units 2nd sem␣
↪(evaluations)',
              'Curricular units 2nd sem (approved)', 'Curricular units 2nd sem␣
↪(grade)',
              'Curricular units 2nd sem (without evaluations)', 'Unemployment␣
↪rate', 'Inflation rate', 'GDP']].describe()
```

```
# Looked at raw data and there are 152 rows with no units for either semester,
↪where the results are almost 50/50 graduate vs dropout.
# Will remove those rows from data set since they are incomplete.
```

[27]:

```
       Age at enrollment  Curricular units 1st sem (credited)  \
count        3630.000000                          3630.000000
mean           23.461157                             0.754270
std             7.827994                             2.477277
min            17.000000                             0.000000
25%            19.000000                             0.000000
50%            20.000000                             0.000000
75%            25.000000                             0.000000
max            70.000000                            20.000000

       Curricular units 1st sem (credited)  \
count                          3630.000000
mean                              0.754270
std                               2.477277
min                               0.000000
25%                               0.000000
50%                               0.000000
75%                               0.000000
max                              20.000000

       Curricular units 1st sem (enrolled)  \
count                          3630.000000
mean                              6.337466
std                               2.570773
min                               0.000000
25%                               5.000000
50%                               6.000000
75%                               7.000000
max                              26.000000

       Curricular units 1st sem (evaluations)  \
count                             3630.000000
mean                                 8.071074
std                                  4.286632
min                                  0.000000
25%                                  6.000000
50%                                  8.000000
75%                                 10.000000
max                                 45.000000

       Curricular units 1st sem (approved)  Curricular units 1st sem (grade)  \
count                          3630.000000                       3630.000000
mean                              4.791460                         10.534860
```

```
std                                 3.237845                           5.057694
min                                 0.000000                           0.000000
25%                                 3.000000                          11.000000
50%                                 5.000000                          12.341429
75%                                 6.000000                          13.500000
max                                26.000000                          18.875000

       Curricular units 1st sem (without evaluations)  \
count                                      3630.000000
mean                                          0.128926
std                                           0.679111
min                                           0.000000
25%                                           0.000000
50%                                           0.000000
75%                                           0.000000
max                                          12.000000

       Curricular units 2nd sem (credited)  \
count                          3630.000000
mean                              0.581818
std                               2.022688
min                               0.000000
25%                               0.000000
50%                               0.000000
75%                               0.000000
max                              19.000000

       Curricular units 2nd sem (enrolled)  \
count                          3630.000000
mean                              6.296419
std                               2.263020
min                               0.000000
25%                               5.000000
50%                               6.000000
75%                               7.000000
max                              23.000000

       Curricular units 2nd sem (evaluations)  \
count                             3630.000000
mean                                 7.763085
std                                  3.964163
min                                  0.000000
25%                                  6.000000
50%                                  8.000000
75%                                 10.000000
max                                 33.000000
```

```
       Curricular units 2nd sem (approved)  Curricular units 2nd sem (grade)  \
count                          3630.000000                       3630.000000
mean                              4.518457                         10.036155
std                               3.162376                          5.481742
min                               0.000000                          0.000000
25%                               2.000000                         10.517857
50%                               5.000000                         12.333333
75%                               6.000000                         13.500000
max                              20.000000                         18.571429

       Curricular units 2nd sem (without evaluations)  Unemployment rate  \
count                                     3630.000000        3630.000000
mean                                         0.142149          11.630358
std                                          0.747670           2.667652
min                                          0.000000           7.600000
25%                                          0.000000           9.400000
50%                                          0.000000          11.100000
75%                                          0.000000          13.900000
max                                         12.000000          16.200000

       Inflation rate          GDP
count     3630.000000  3630.000000
mean         1.231598    -0.009256
std          1.384911     2.259986
min         -0.800000    -4.060000
25%          0.300000    -1.700000
50%          1.400000     0.320000
75%          2.600000     1.790000
max          3.700000     3.510000
```

[29]:
```
# Removing no credit hour rows
students_data = students_data[students_data['Curricular units 1st sem␣
 ↪(enrolled)'] !=0]
students_data.info()
# Removed 152 rows
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 3478 entries, 1 to 4423
Data columns (total 36 columns):
 #   Column                                          Non-Null Count  Dtype
---  ------                                          --------------  -----
 0   Marital status                                  3478 non-null   int64
 1   Application mode                                3478 non-null   int64
 2   Application order                               3478 non-null   int64
 3   Course                                          3478 non-null   int64
 4   Attendance Time                                 3478 non-null   int64
 5   Previous qualification                          3478 non-null   int64
 6   Nacionality                                     3478 non-null   int64
```

```
 7   Mom qualification                             3478 non-null   int64
 8   Dad qualification                             3478 non-null   int64
 9   Mom occupation                                3478 non-null   int64
 10  Dad occupation                                3478 non-null   int64
 11  Displaced                                     3478 non-null   int64
 12  Educational special needs                     3478 non-null   int64
 13  Debtor                                        3478 non-null   int64
 14  Tuition fees up to date                       3478 non-null   int64
 15  Gender                                        3478 non-null   int64
 16  Scholarship holder                            3478 non-null   int64
 17  Age at enrollment                             3478 non-null   int64
 18  International                                 3478 non-null   int64
 19  Curricular units 1st sem (credited)           3478 non-null   int64
 20  Curricular units 1st sem (enrolled)           3478 non-null   int64
 21  Curricular units 1st sem (evaluations)        3478 non-null   int64
 22  Curricular units 1st sem (approved)           3478 non-null   int64
 23  Curricular units 1st sem (grade)              3478 non-null   float64
 24  Curricular units 1st sem (without evaluations) 3478 non-null   int64
 25  Curricular units 2nd sem (credited)           3478 non-null   int64
 26  Curricular units 2nd sem (enrolled)           3478 non-null   int64
 27  Curricular units 2nd sem (evaluations)        3478 non-null   int64
 28  Curricular units 2nd sem (approved)           3478 non-null   int64
 29  Curricular units 2nd sem (grade)              3478 non-null   float64
 30  Curricular units 2nd sem (without evaluations) 3478 non-null   int64
 31  Unemployment rate                             3478 non-null   float64
 32  Inflation rate                                3478 non-null   float64
 33  GDP                                           3478 non-null   float64
 34  Target                                        3478 non-null   object
 35  Target_Int                                    3478 non-null   int64
dtypes: float64(5), int64(30), object(1)
memory usage: 1005.4+ KB
```

[31]:
```
# Rerunning statistics
students_data[['Age at enrollment', 'Curricular units 1st sem (credited)',␣
↪'Curricular units 1st sem (credited)',
               'Curricular units 1st sem (enrolled)', 'Curricular units 1st sem␣
↪(evaluations)',
               'Curricular units 1st sem (approved)', 'Curricular units 1st sem␣
↪(grade)',
               'Curricular units 1st sem (without evaluations)', 'Curricular␣
↪units 2nd sem (credited)',
               'Curricular units 2nd sem (enrolled)', 'Curricular units 2nd sem␣
↪(evaluations)',
               'Curricular units 2nd sem (approved)', 'Curricular units 2nd sem␣
↪(grade)',
               'Curricular units 2nd sem (without evaluations)', 'Unemployment␣
↪rate', 'Inflation rate', 'GDP']].describe()
```

[31]:

```
       Age at enrollment  Curricular units 1st sem (credited)  \
count        3478.000000                          3478.000000
mean           23.596895                             0.787234
std             7.930037                             2.525712
min            17.000000                             0.000000
25%            19.000000                             0.000000
50%            20.000000                             0.000000
75%            26.000000                             0.000000
max            70.000000                            20.000000

       Curricular units 1st sem (credited)  \
count                          3478.000000
mean                              0.787234
std                               2.525712
min                               0.000000
25%                               0.000000
50%                               0.000000
75%                               0.000000
max                              20.000000

       Curricular units 1st sem (enrolled)  \
count                          3478.000000
mean                              6.614434
std                               2.250616
min                               1.000000
25%                               6.000000
50%                               6.000000
75%                               7.000000
max                              26.000000

       Curricular units 1st sem (evaluations)  \
count                             3478.000000
mean                                 8.423807
std                                  4.025705
min                                  0.000000
25%                                  6.000000
50%                                  8.000000
75%                                 10.000000
max                                 45.000000

       Curricular units 1st sem (approved)  Curricular units 1st sem (grade)  \
count                          3478.000000                       3478.000000
mean                              5.000863                         10.995268
std                               3.145544                          4.651319
min                               0.000000                          0.000000
25%                               3.000000                         11.295833
50%                               6.000000                         12.500000
```

```
75%                                6.000000                          13.500000
max                               26.000000                          18.875000

       Curricular units 1st sem (without evaluations)  \
count                                     3478.000000
mean                                         0.134560
std                                          0.693249
min                                          0.000000
25%                                          0.000000
50%                                          0.000000
75%                                          0.000000
max                                         12.000000

       Curricular units 2nd sem (credited)  \
count                          3478.000000
mean                              0.607246
std                               2.062686
min                               0.000000
25%                               0.000000
50%                               0.000000
75%                               0.000000
max                              19.000000

       Curricular units 2nd sem (enrolled)  \
count                          3478.000000
mean                              6.571593
std                               1.880502
min                               1.000000
25%                               6.000000
50%                               6.000000
75%                               7.000000
max                              23.000000

       Curricular units 2nd sem (evaluations)  \
count                             3478.000000
mean                                 8.102358
std                                  3.694844
min                                  0.000000
25%                                  6.000000
50%                                  8.000000
75%                                 10.000000
max                                 33.000000

       Curricular units 2nd sem (approved)  Curricular units 2nd sem (grade)  \
count                          3478.000000                       3478.000000
mean                              4.715929                         10.474768
std                               3.083226                          5.173727
```

```
min                                   0.000000                            0.000000
25%                                   3.000000                           11.000000
50%                                   5.000000                           12.400000
75%                                   6.000000                           13.540208
max                                  20.000000                           18.571429

       Curricular units 2nd sem (without evaluations)  Unemployment rate  \
count                                     3478.000000        3478.000000
mean                                         0.148361          11.627717
std                                          0.763234           2.669662
min                                          0.000000           7.600000
25%                                          0.000000           9.400000
50%                                          0.000000          11.100000
75%                                          0.000000          13.900000
max                                         12.000000          16.200000

       Inflation rate          GDP
count     3478.000000  3478.000000
mean         1.235106     0.008856
std          1.379361     2.263506
min         -0.800000    -4.060000
25%          0.300000    -1.700000
50%          1.400000     0.320000
75%          2.600000     1.790000
max          3.700000     3.510000
```

[33]:

```python
# Apply the school grouping used during graphing to the full data set.
students_data['Mom School'] = students_data['Mom qualification'].
 ↪apply(school_type)
students_data['Dad School'] = students_data['Dad qualification'].
 ↪apply(school_type)
# Remove original columns
students_data = students_data.drop(['Dad qualification', 'Mom qualification'],␣
 ↪axis=1)
students_data.head()
```

[33]:

```
   Marital status  Application mode  Application order  Course  \
1               1                 6                  1      11
2               1                 1                  5       5
3               1                 8                  2      15
4               2                12                  1       3
5               2                12                  1      17

   Attendance Time  Previous qualification  Nacionality  Mom occupation  \
1                1                       1            1               4
2                1                       1            1              10
3                1                       1            1               6
```

```
4              0                      1            1             10
5              0                     12            1             10

   Dad occupation  Displaced  …  Curricular units 2nd sem (approved)  \
1               4          1  …                                    6
2              10          1  …                                    0
3               4          1  …                                    5
4              10          0  …                                    6
5               8          0  …                                    5

   Curricular units 2nd sem (grade)  \
1                         13.666667
2                          0.000000
3                         12.400000
4                         13.000000
5                         11.500000

   Curricular units 2nd sem (without evaluations)  Unemployment rate  \
1                                               0               13.9
2                                               0               10.8
3                                               0                9.4
4                                               0               13.9
5                                               5               16.2

   Inflation rate   GDP    Target  Target_Int     Mom School       Dad School
1            -0.3  0.79  Graduate           1  3-High School        5-College
2             1.4  1.74   Dropout           0    4-Associate     1-Elementary
3            -0.8 -3.12  Graduate           1    4-Associate     1-Elementary
4            -0.3  0.79  Graduate           1    4-Associate  2-Middle School
5             0.3 -0.92  Graduate           1    4-Associate     1-Elementary

[5 rows x 36 columns]
```

[35]:

```python
# Create functions to update categorical variables to provide the value not a␣
↪number

def yesno(x):
    if x in [0]:
        return 'no'
    elif x in [1]:
        return 'yes'
    else:
        return 'other'

def gender(x):
    if x in [0]:
        return 'female'
```

```
    elif x in [1]:
        return 'male'
    else:
        return 'other'

def attendance(x):
    if x in [0]:
        return 'evening'
    elif x in [1]:
        return 'daytime'
    else:
        return 'other'

def marital_status(x):
    if x in [1]:
        return 'single'
    elif x in [2]:
        return 'married'
    elif x in [3]:
        return 'widower'
    elif x in [4]:
        return 'divorced'
    elif x in [5,6]:
        return 'other'
    else:
        return 'other'

def nacionality(x):
    if x in [1]:
        return 'Portuguese'
    elif x in [3]:
        return 'Spanish'
    elif x in [9]:
        return 'Cape Verdean'
    elif x in [12]:
        return 'Santomean'
    elif x in [14]:
        return 'Braziliam'
    else:
        return 'Other'

def course(x):
    if x in [1]:
        return 'Biofuel Production Technologies'
    elif x in [2]:
        return 'Animation and Multimedia Design'
    elif x in [3,10]:
```

```
        return 'Social Service'
    elif x in [4]:
        return 'Agronomy'
    elif x in [5]:
        return 'Communication Design'
    elif x in [6]:
        return 'Veterinary Nursing'
    elif x in [7]:
        return 'Informatics Engineering'
    elif x in [8]:
        return 'Equiniculture'
    elif x in [9,17]:
        return 'Management'
    elif x in [11]:
        return 'Tourism'
    elif x in [12]:
        return 'Nursing'
    elif x in [13]:
        return 'Oral Hygiene'
    elif x in [14]:
        return 'Advertising and Marketing Management'
    elif x in [15]:
        return 'Journalism and Communication'
    elif x in [16]:
        return 'Basic Education'
    else:
        return 'other'

def prev_qual(x):
    if x in [13]:
        return '2-Middle School'
    elif x in [1, 7, 8, 9, 10, 11, 12, 14,]:
        return '3-High School'
    elif x in [16, ]:
        return '4-Associate'
    elif x in [2, 3, 4, 5, 6, 15, 17 ]:
        return '5-College'
    else:
        return 'Other'

def occupation(x):
    if x in [5, 22, 27, 28, 29]:
        return 'Adminitrative Support'
    elif x in [7, 34, 35, 43]:
        return 'Agriculture'
    elif x in [3, 19, 21]:
        return 'Education'
```

```
    elif x in [2,17]:
        return 'Executive'
    elif x in [20,24]:
        return 'Healthcare'
    elif x in [4, 23, 25, 26]:
        return 'Information Technology'
    elif x in [11, 14, 15, 16]:
        return 'Military'
    elif x in [6, 30, 32, 33]:
        return 'Personal Services'
    elif x in [18, 39, 45, 46]:
        return 'Service Industry'
    elif x in [1]:
        return 'Student'
    elif x in [40, 41, 42]:
        return 'Technical'
    elif x in [8, 9, 31, 36, 37, 38]:
        return 'Trades'
    else:
        return 'other'
```

[37]:

```
students_data['Displaced'] = students_data['Displaced'].apply(yesno)
students_data['Educational special needs'] = students_data['Educational special needs'].apply(yesno)
students_data['Debtor'] = students_data['Debtor'].apply(yesno)
students_data['Tuition fees up to date'] = students_data['Tuition fees up to date'].apply(yesno)
students_data['Scholarship holder'] = students_data['Scholarship holder'].apply(yesno)
students_data['International'] = students_data['International'].apply(yesno)
students_data['Gender'] = students_data['Gender'].apply(gender)
students_data['Attendance Time'] = students_data['Attendance Time'].apply(attendance)
students_data['Marital status'] = students_data['Marital status'].apply(marital_status)
students_data['Nacionality'] = students_data['Nacionality'].apply(nacionality)
students_data['Course'] = students_data['Course'].apply(course)
students_data['Previous qualification'] = students_data['Previous qualification'].apply(prev_qual)
students_data['Mom occupation'] = students_data['Mom occupation'].apply(occupation)
students_data['Dad occupation'] = students_data['Dad occupation'].apply(occupation)
students_data.head()
```

[37]:
```
  Marital status  Application mode  Application order  \
1         single                 6                  1
2         single                 1                  5
3         single                 8                  2
4        married                12                  1
5        married                12                  1

                      Course Attendance Time Previous qualification  \
1                    Tourism         daytime          3-High School
2       Communication Design         daytime          3-High School
3  Journalism and Communication      daytime          3-High School
4             Social Service         evening          3-High School
5                 Management         evening          3-High School

  Nacionality          Mom occupation          Dad occupation Displaced  …  \
1  Portuguese  Information Technology  Information Technology       yes  …
2  Portuguese                   other                   other       yes  …
3  Portuguese       Personal Services  Information Technology       yes  …
4  Portuguese                   other                   other        no  …
5  Portuguese                   other                  Trades        no  …

  Curricular units 2nd sem (approved) Curricular units 2nd sem (grade)  \
1                                   6                        13.666667
2                                   0                         0.000000
3                                   5                        12.400000
4                                   6                        13.000000
5                                   5                        11.500000

  Curricular units 2nd sem (without evaluations) Unemployment rate  \
1                                              0              13.9
2                                              0              10.8
3                                              0               9.4
4                                              0              13.9
5                                              5              16.2

  Inflation rate   GDP    Target  Target_Int     Mom School        Dad School
1           -0.3  0.79  Graduate           1  3-High School         5-College
2            1.4  1.74   Dropout           0    4-Associate      1-Elementary
3           -0.8 -3.12  Graduate           1    4-Associate      1-Elementary
4           -0.3  0.79  Graduate           1    4-Associate  2-Middle School
5            0.3 -0.92  Graduate           1    4-Associate      1-Elementary

[5 rows x 36 columns]
```

[39]:
```
# Remove Application Mode because no dictionary to go with values and they must be international based as I can't understand them
```

```
# Remove Application order because the order in which they apply for classes as␣
↪it relates to the student it unknown because there isn't a student ID
# Remove Target field as Target ID is the true target variable
students_data = students_data.drop(['Application mode', 'Application␣
↪order','Target'], axis=1)
students_data.head()
```

[39]:

```
  Marital status                        Course Attendance Time  \
1         single                       Tourism         daytime
2         single          Communication Design         daytime
3         single  Journalism and Communication         daytime
4        married                Social Service         evening
5        married                    Management         evening

  Previous qualification Nacionality          Mom occupation  \
1          3-High School  Portuguese  Information Technology
2          3-High School  Portuguese                   other
3          3-High School  Portuguese       Personal Services
4          3-High School  Portuguese                   other
5          3-High School  Portuguese                   other

           Dad occupation Displaced Educational special needs Debtor  …  \
1  Information Technology       yes                       no     no  …
2                   other       yes                       no     no  …
3  Information Technology       yes                       no     no  …
4                   other        no                       no     no  …
5                  Trades        no                       no    yes  …

  Curricular units 2nd sem (evaluations) Curricular units 2nd sem (approved)  \
1                                      6                                   6
2                                      0                                   0
3                                     10                                   5
4                                      6                                   6
5                                     17                                   5

  Curricular units 2nd sem (grade)  \
1                        13.666667
2                         0.000000
3                        12.400000
4                        13.000000
5                        11.500000

   Curricular units 2nd sem (without evaluations) Unemployment rate  \
1                                               0              13.9
2                                               0              10.8
3                                               0               9.4
4                                               0              13.9
```

```
5                                               5             16.2

   Inflation rate   GDP  Target_Int     Mom School       Dad School
1            -0.3  0.79           1  3-High School        5-College
2             1.4  1.74           0    4-Associate     1-Elementary
3            -0.8 -3.12           1    4-Associate     1-Elementary
4            -0.3  0.79           1    4-Associate  2-Middle School
5             0.3 -0.92           1    4-Associate     1-Elementary

[5 rows x 33 columns]
```

[41]:

```
students_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 3478 entries, 1 to 4423
Data columns (total 33 columns):
 #   Column                                         Non-Null Count  Dtype
---  ------                                         --------------  -----
 0   Marital status                                 3478 non-null   object
 1   Course                                         3478 non-null   object
 2   Attendance Time                                3478 non-null   object
 3   Previous qualification                         3478 non-null   object
 4   Nacionality                                    3478 non-null   object
 5   Mom occupation                                 3478 non-null   object
 6   Dad occupation                                 3478 non-null   object
 7   Displaced                                      3478 non-null   object
 8   Educational special needs                      3478 non-null   object
 9   Debtor                                         3478 non-null   object
 10  Tuition fees up to date                        3478 non-null   object
 11  Gender                                         3478 non-null   object
 12  Scholarship holder                             3478 non-null   object
 13  Age at enrollment                              3478 non-null   int64
 14  International                                  3478 non-null   object
 15  Curricular units 1st sem (credited)            3478 non-null   int64
 16  Curricular units 1st sem (enrolled)            3478 non-null   int64
 17  Curricular units 1st sem (evaluations)         3478 non-null   int64
 18  Curricular units 1st sem (approved)            3478 non-null   int64
 19  Curricular units 1st sem (grade)               3478 non-null   float64
 20  Curricular units 1st sem (without evaluations) 3478 non-null   int64
 21  Curricular units 2nd sem (credited)            3478 non-null   int64
 22  Curricular units 2nd sem (enrolled)            3478 non-null   int64
 23  Curricular units 2nd sem (evaluations)         3478 non-null   int64
 24  Curricular units 2nd sem (approved)            3478 non-null   int64
 25  Curricular units 2nd sem (grade)               3478 non-null   float64
 26  Curricular units 2nd sem (without evaluations) 3478 non-null   int64
 27  Unemployment rate                              3478 non-null   float64
 28  Inflation rate                                 3478 non-null   float64
 29  GDP                                            3478 non-null   float64
```

```
 30   Target_Int                                  3478 non-null   int64
 31   Mom School                                  3478 non-null   object
 32   Dad School                                  3478 non-null   object
dtypes: float64(5), int64(12), object(16)
memory usage: 923.8+ KB
```

[43]:
```python
# Convert the categorical columns to dummy variables.
# Note - categorial fields were all assigned numbers with a reference/lookup
 ↪table as to the actual values but still need to dummy them out for modeling
 ↪because the numbers are not related.
cat_columns = ['Marital status', 'Course', 'Attendance Time', 'Previous
 ↪qualification', 'Nacionality', 'Mom occupation', 'Dad occupation', 'Mom
 ↪School', 'Dad School',
               'Displaced', 'Educational special needs', 'Debtor', 'Tuition
 ↪fees up to date', 'Gender', 'Scholarship holder', 'International']
students_encoded = pd.get_dummies(students_data, columns=cat_columns)
students_encoded.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 3478 entries, 1 to 4423
Data columns (total 98 columns):
 #    Column                                      Non-Null Count  Dtype
---   ------                                      --------------  -----
 0    Age at enrollment                           3478 non-null   int64
 1    Curricular units 1st sem (credited)         3478 non-null   int64
 2    Curricular units 1st sem (enrolled)         3478 non-null   int64
 3    Curricular units 1st sem (evaluations)      3478 non-null   int64
 4    Curricular units 1st sem (approved)         3478 non-null   int64
 5    Curricular units 1st sem (grade)            3478 non-null   float64
 6    Curricular units 1st sem (without evaluations) 3478 non-null   int64
 7    Curricular units 2nd sem (credited)         3478 non-null   int64
 8    Curricular units 2nd sem (enrolled)         3478 non-null   int64
 9    Curricular units 2nd sem (evaluations)      3478 non-null   int64
 10   Curricular units 2nd sem (approved)         3478 non-null   int64
 11   Curricular units 2nd sem (grade)            3478 non-null   float64
 12   Curricular units 2nd sem (without evaluations) 3478 non-null   int64
 13   Unemployment rate                           3478 non-null   float64
 14   Inflation rate                              3478 non-null   float64
 15   GDP                                         3478 non-null   float64
 16   Target_Int                                  3478 non-null   int64
 17   Marital status_divorced                     3478 non-null   bool
 18   Marital status_married                      3478 non-null   bool
 19   Marital status_other                        3478 non-null   bool
 20   Marital status_single                       3478 non-null   bool
 21   Marital status_widower                      3478 non-null   bool
 22   Course_Advertising and Marketing Management 3478 non-null   bool
 23   Course_Agronomy                             3478 non-null   bool
 24   Course_Animation and Multimedia Design      3478 non-null   bool
```

```
25  Course_Basic Education                        3478 non-null   bool
26  Course_Biofuel Production Technologies        3478 non-null   bool
27  Course_Communication Design                   3478 non-null   bool
28  Course_Equiniculture                          3478 non-null   bool
29  Course_Informatics Engineering                3478 non-null   bool
30  Course_Journalism and Communication           3478 non-null   bool
31  Course_Management                             3478 non-null   bool
32  Course_Nursing                                3478 non-null   bool
33  Course_Oral Hygiene                           3478 non-null   bool
34  Course_Social Service                         3478 non-null   bool
35  Course_Tourism                                3478 non-null   bool
36  Course_Veterinary Nursing                     3478 non-null   bool
37  Attendance Time_daytime                       3478 non-null   bool
38  Attendance Time_evening                       3478 non-null   bool
39  Previous qualification_2-Middle School        3478 non-null   bool
40  Previous qualification_3-High School          3478 non-null   bool
41  Previous qualification_4-Associate            3478 non-null   bool
42  Previous qualification_5-College              3478 non-null   bool
43  Nacionality_Braziliam                         3478 non-null   bool
44  Nacionality_Cape Verdean                      3478 non-null   bool
45  Nacionality_Other                             3478 non-null   bool
46  Nacionality_Portuguese                        3478 non-null   bool
47  Nacionality_Santomean                         3478 non-null   bool
48  Nacionality_Spanish                           3478 non-null   bool
49  Mom occupation_Adminitrative Support          3478 non-null   bool
50  Mom occupation_Agriculture                    3478 non-null   bool
51  Mom occupation_Education                      3478 non-null   bool
52  Mom occupation_Executive                      3478 non-null   bool
53  Mom occupation_Healthcare                     3478 non-null   bool
54  Mom occupation_Information Technology         3478 non-null   bool
55  Mom occupation_Military                       3478 non-null   bool
56  Mom occupation_Personal Services              3478 non-null   bool
57  Mom occupation_Service Industry               3478 non-null   bool
58  Mom occupation_Student                        3478 non-null   bool
59  Mom occupation_Trades                         3478 non-null   bool
60  Mom occupation_other                          3478 non-null   bool
61  Dad occupation_Adminitrative Support          3478 non-null   bool
62  Dad occupation_Agriculture                    3478 non-null   bool
63  Dad occupation_Education                      3478 non-null   bool
64  Dad occupation_Executive                      3478 non-null   bool
65  Dad occupation_Healthcare                     3478 non-null   bool
66  Dad occupation_Information Technology         3478 non-null   bool
67  Dad occupation_Military                       3478 non-null   bool
68  Dad occupation_Personal Services              3478 non-null   bool
69  Dad occupation_Service Industry               3478 non-null   bool
70  Dad occupation_Student                        3478 non-null   bool
71  Dad occupation_Technical                      3478 non-null   bool
72  Dad occupation_Trades                         3478 non-null   bool
```

```
 73  Dad occupation_other                              3478 non-null   bool
 74  Mom School_1-Elementary                           3478 non-null   bool
 75  Mom School_2-Middle School                        3478 non-null   bool
 76  Mom School_3-High School                          3478 non-null   bool
 77  Mom School_4-Associate                            3478 non-null   bool
 78  Mom School_5-College                              3478 non-null   bool
 79  Dad School_1-Elementary                           3478 non-null   bool
 80  Dad School_2-Middle School                        3478 non-null   bool
 81  Dad School_3-High School                          3478 non-null   bool
 82  Dad School_4-Associate                            3478 non-null   bool
 83  Dad School_5-College                              3478 non-null   bool
 84  Displaced_no                                      3478 non-null   bool
 85  Displaced_yes                                     3478 non-null   bool
 86  Educational special needs_no                      3478 non-null   bool
 87  Educational special needs_yes                     3478 non-null   bool
 88  Debtor_no                                         3478 non-null   bool
 89  Debtor_yes                                        3478 non-null   bool
 90  Tuition fees up to date_no                        3478 non-null   bool
 91  Tuition fees up to date_yes                       3478 non-null   bool
 92  Gender_female                                     3478 non-null   bool
 93  Gender_male                                       3478 non-null   bool
 94  Scholarship holder_no                             3478 non-null   bool
 95  Scholarship holder_yes                            3478 non-null   bool
 96  International_no                                  3478 non-null   bool
 97  International_yes                                 3478 non-null   bool
dtypes: bool(81), float64(5), int64(12)
memory usage: 764.2 KB
```

[45]:
```
# Perform feature reduction through PCA
from sklearn.decomposition import PCA
```

[47]:
```
# Set PCA to retain variables that 98% of the variance is retained
pca = PCA(n_components=0.98, whiten=True)
students_pca = pca.fit_transform(students_encoded)
```

[49]:
```
# How many features are in the PCA-transformed matrix?
print("Original number of features:", students_encoded.shape[1])
print("Reduced number of features:", students_pca.shape[1])
```

```
Original number of features: 98
Reduced number of features: 19
```